Task 1:

LAN A:

- Subnet Mask: 255.255.255.192 /26
- Network Address: 20.10.172.128
- Smallest IP Address: 20.10.172.129
- Highest IP Address: 20.10.172.190

LAN B:

- Subnet Mask: 255.255.255.128 /25
- Network Address: 20.10.172.0
- Smallest IP Address: 20.10.172.1
- Highest IP Address: 20.10.172.126

LAN C:

- Subnet Mask: 255.255.255.224 /27
- Network Address: 20.10.172.192
- Smallest IP Address: 20.10.172.193
- Highest IP Address: 20.10.172.222

Output for Task 2:

```
mininet@mininet-vm:/media/sf_Data-Comm-Hw5/Data-Comm-Hw5$ sudo python3 layer3_network_code.py
*** Configuring hosts
rA rB rC hA1 hA2 hB1 hB2 hC1 hC2
*** Starting controller

*** Starting 3 switches
s1 s2 s3 ...

Testing LAN connectivity:

LAN A:
hA1 -> hA2
hA2 -> hA1
*** Results: 0% dropped (2/2 received)
0.0
LAN B:
hB1 -> hB2
hB2 -> hB1
*** Results: 0% dropped (2/2 received)
0.0
LAN C:
hC1 -> hC2
hC2 -> hC1
*** Results: 0% dropped (2/2 received)
0.0
```

```
Cross-LAN test (shouldn't work in Task 2):

*** Ping: testing ping reachability
rA -> X X hA1 hA2 X X X X
rB -> X X X X hB1 hB2 X X
rC -> X X X X X X hC1 hC2
hA1 -> rA X X hA2 X X X X
hA2 -> rA X X hA1 X X X X
hB1 -> X rB X X X hB2 X X
hB2 -> X rB X X X hB1 X X
hC1 -> X X rC X X X X hC2
hC2 -> X X rC X X X X hC1
*** Results: 75% dropped (18/72 received)
75.0
*** Starting CLI:
```

mininet> rA route add -net 20.10.172.0 netmask 255.255.255.128 gw 20.10.100.2
mininet> rA route add -net 20.10.172.192 netmask 255.255.255.224 gw 20.10.100.3
mininet> rB route add -net 20.10.172.128 netmask 255.255.255.192 gw 20.10.100.1
mininet> rB route add -net 20.10.172.192 netmask 255.255.255.224 gw 20.10.100.3
mininet> rC route add -net 20.10.172.128 netmask 255.255.255.192 gw 20.10.100.1
mininet> rC route add -net 20.10.172.0 netmask 255.255.255.128 gw 20.10.100.2

Routing Information:

```
mininet> rA sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
mininet> rB sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
mininet> rC sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
mininet> rA route add -net 20.10.172.0 netmask 255.255.255.128 gw 20.10.100.2
mininet> rA route add -net 20.10.172.192 netmask 255.255.255.224 gw 20.10.100.3
mininet> rB route add -net 20.10.172.128 netmask 255.255.255.192 gw 20.10.100.1
mininet> rB route add -net 20.10.172.192 netmask 255.255.255.224 gw 20.10.100.3
mininet> rC route add -net 20.10.172.128 netmask 255.255.255.192 gw 20.10.100.1
mininet> rC route add -net 20.10.172.0 netmask 255.255.255.128 gw 20.10.100.2
```

Output for Task 3:

- Ping examples:

```
mininet> hA1 ping -c 4 hB1
PING 20.10.172.10 (20.10.172.10) 56(84) bytes of data.
64 bytes from 20.10.172.10: icmp_seq=1 ttl=62 time=0.759 ms
64 bytes from 20.10.172.10: icmp_seq=2 ttl=62 time=0.244 ms
64 bytes from 20.10.172.10: icmp_seq=3 ttl=62 time=0.076 ms
64 bytes from 20.10.172.10: icmp_seq=4 ttl=62 time=0.090 ms

--- 20.10.172.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3038ms
rtt min/avg/max/mdev = 0.076/0.292/0.759/0.277 ms
```

- Traceroute examples:

```
mininet> hA1 traceroute hC1
traceroute to 20.10.172.193 (20.10.172.193), 30 hops max, 60 byte packets
 1  20.10.172.129 (20.10.172.129)  0.069 ms  0.016 ms  0.014 ms
 2  20.10.100.3 (20.10.100.3)  0.625 ms  0.037 ms  0.021 ms
 3  20.10.172.193 (20.10.172.193)  0.269 ms  0.035 ms  0.027 ms
mininet> hA1 traceroute hB1
traceroute to 20.10.172.10 (20.10.172.10), 30 hops max, 60 byte packets
 1  20.10.172.129 (20.10.172.129)  0.069 ms  0.018 ms  0.013 ms
 2  20.10.100.2 (20.10.100.2)  0.291 ms  0.031 ms  0.020 ms
 3  20.10.172.10 (20.10.172.10)  0.169 ms  0.031 ms  0.024 ms
mininet> hB1 traceroute hC1
traceroute to 20.10.172.193 (20.10.172.193), 30 hops max, 60 byte packets
 1  20.10.172.1 (20.10.172.1)  0.071 ms  0.017 ms  0.014 ms
 2  20.10.100.3 (20.10.100.3)  0.439 ms  0.069 ms  0.022 ms
 3  20.10.172.193 (20.10.172.193)  0.318 ms  0.044 ms  0.026 ms
mininet> hA2 traceroute hC1
traceroute to 20.10.172.193 (20.10.172.193), 30 hops max, 60 byte packets
 1  20.10.172.129 (20.10.172.129)  22.620 ms  30.753 ms  30.789 ms
 2  20.10.100.3 (20.10.100.3)  31.999 ms  32.516 ms  32.478 ms
 3  20.10.172.193 (20.10.172.193)  32.948 ms  32.939 ms  32.861 ms
```